



Machine Learning-Based Human Skill Estimation from Robot Kinematic Data

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Abstract

Shared-autonomy robots require estimates of user skill to adjust assistance appropriately. This study evaluates whether handcrafted, task-independent robot kinematic features can predict continuous skill scores. Speed, acceleration, jerk, idle behavior, and duration features were extracted from 103 JIGSAWS demonstrations and used to train regression models. Leave-One-Subject-Out cross-validation was used to test generalization to unseen users. Tuned Gradient Boosting achieved the best performance with an **MAE of 4.30** GRS points and an **R^2 of 0.13**, outperforming the Dummy Regressor by approximately **12%**. The results indicate that the selected features contain measurable skill-related information, but are insufficient for accurate prediction across the full skill range.

Introduction

Shared autonomy allows robots to adapt assistance while preserving human control. A key requirement is estimating user skill continuously during task execution.

Existing approaches often rely on:

- Task-specific performance metrics
- Manual expert assessment
- Discrete skill classifications

These approaches may not generalize well to new manipulation tasks.

Research Question:

Can handcrafted, task-independent robot kinematic features accurately estimate continuous human skill?

Methodology

Dataset

- JIGSAWS dataset
- 103 demonstrations
- 8 participants
- Knot Tying, Needle Passing, and Suturing
- 30 Hz robot kinematics
- GRS score from 0–30 used as the target

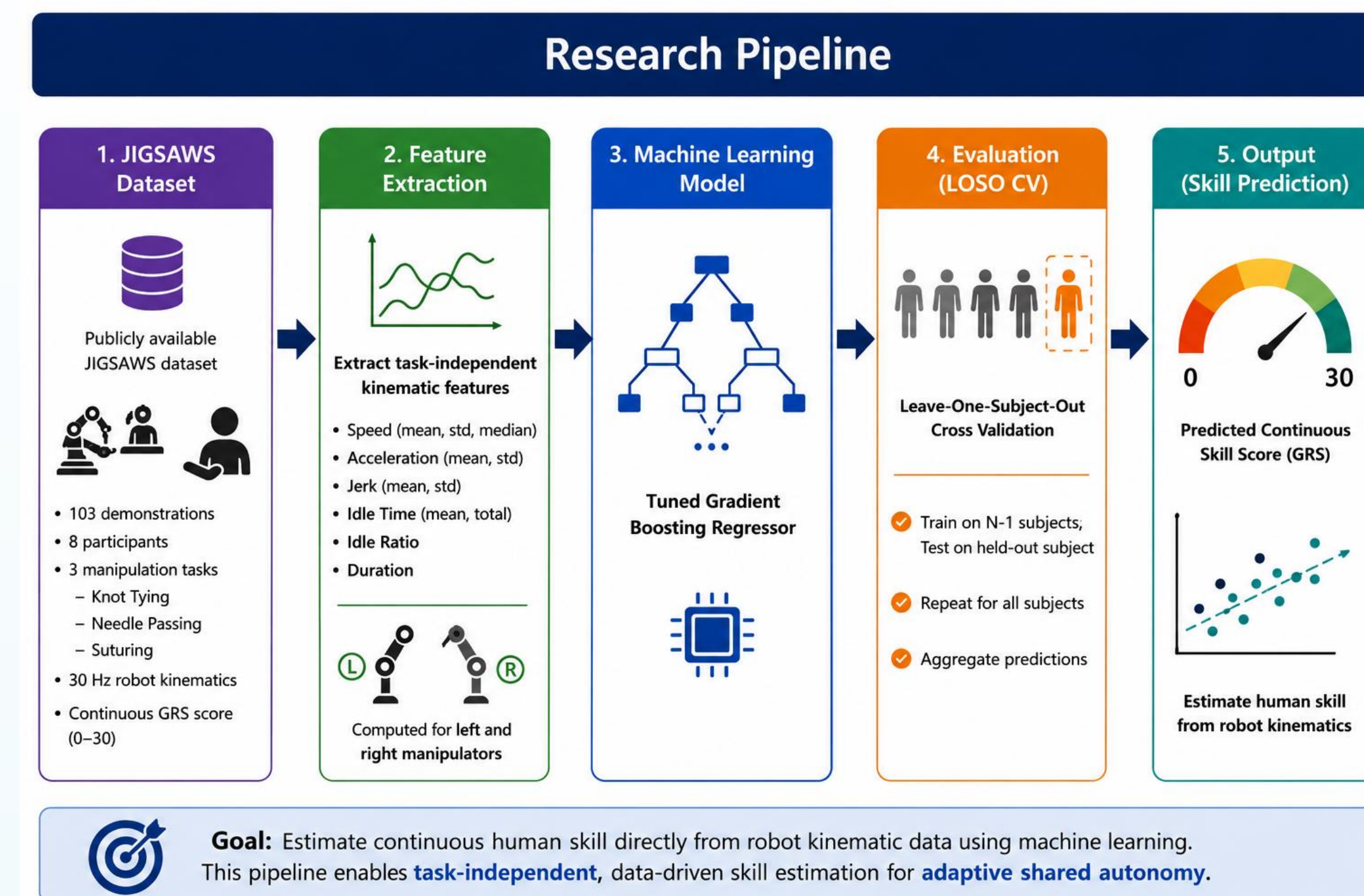
Extracted Features

- Speed
- Acceleration
- Jerk
- Idle time
- Idle ratio
- Task duration

Features were computed separately for the left and right manipulators.

Modeling and Evaluation

- Gradient Boosting Regressor
- Leave-One-Subject-Out cross-validation
- Nested hyperparameter tuning
- Metrics: MAE, RMSE, and R^2



Results

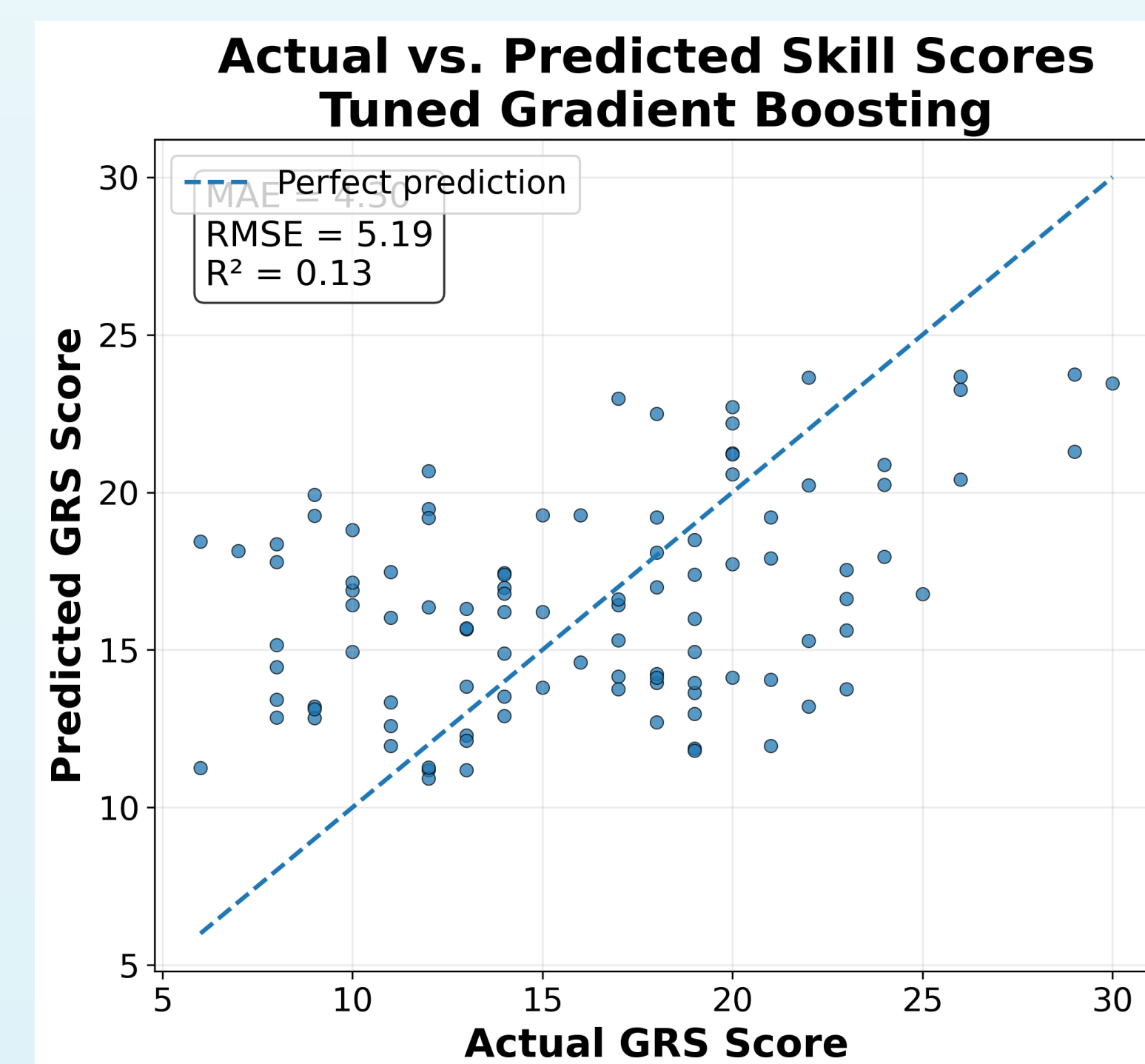
Overall Model Performance

Tuned Gradient Boosting

- MAE = **4.30**
- RMSE = **5.19**
- R^2 = **0.13**

MAE reduced by approximately **12%** compared with the Dummy Regressor

Main finding: The selected features contained predictive signal, but overall skill-estimation accuracy remained limited.

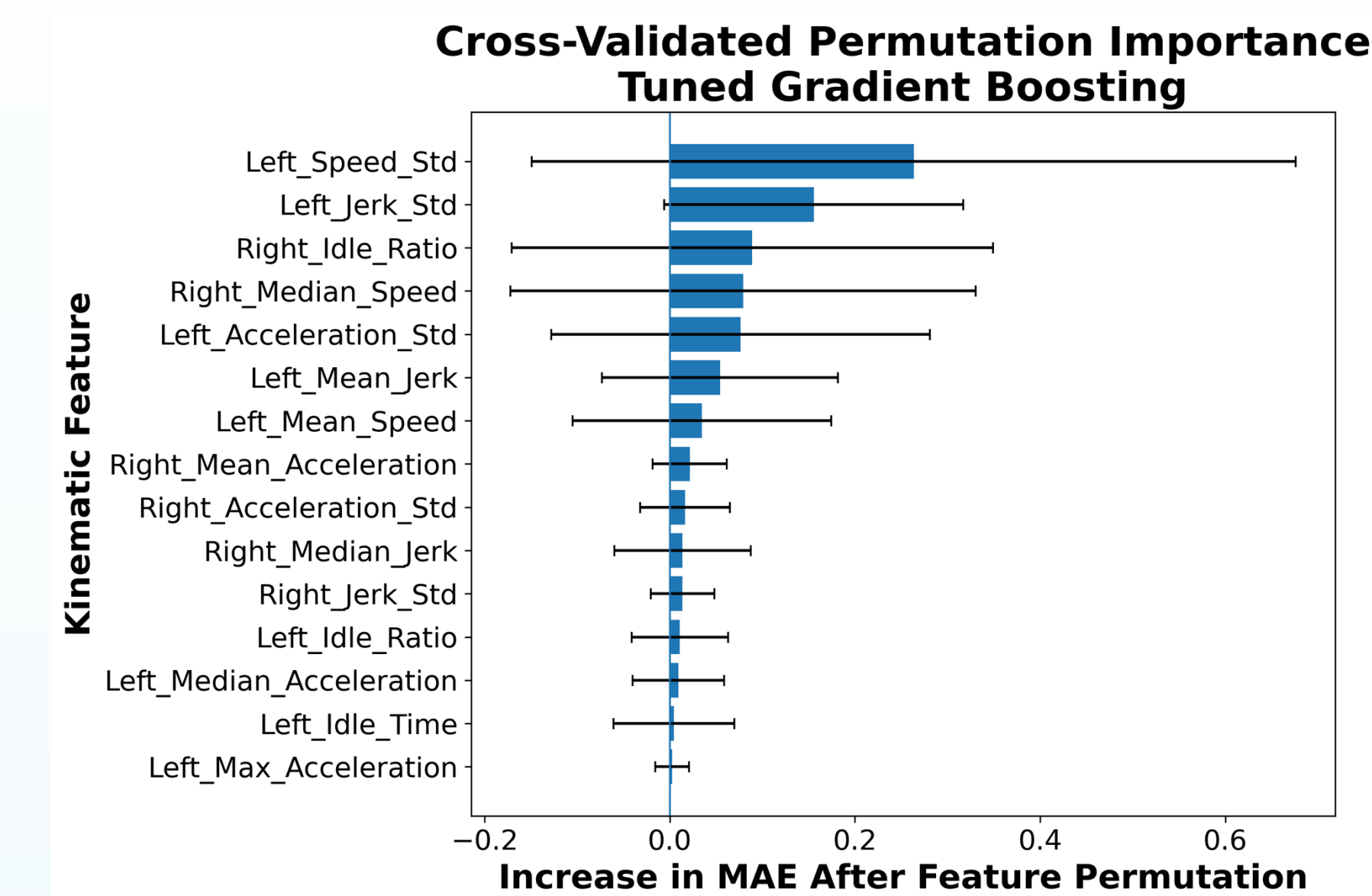


Feature Importance

Permutation importance suggested that speed variability and movement smoothness contributed most strongly to prediction. Most informative individual features included:

- Left speed variability
- Left jerk variability
- Right idle ratio
- Right median speed
- Left acceleration variability

Left jerk variability was the most consistent predictor across held-out subjects.



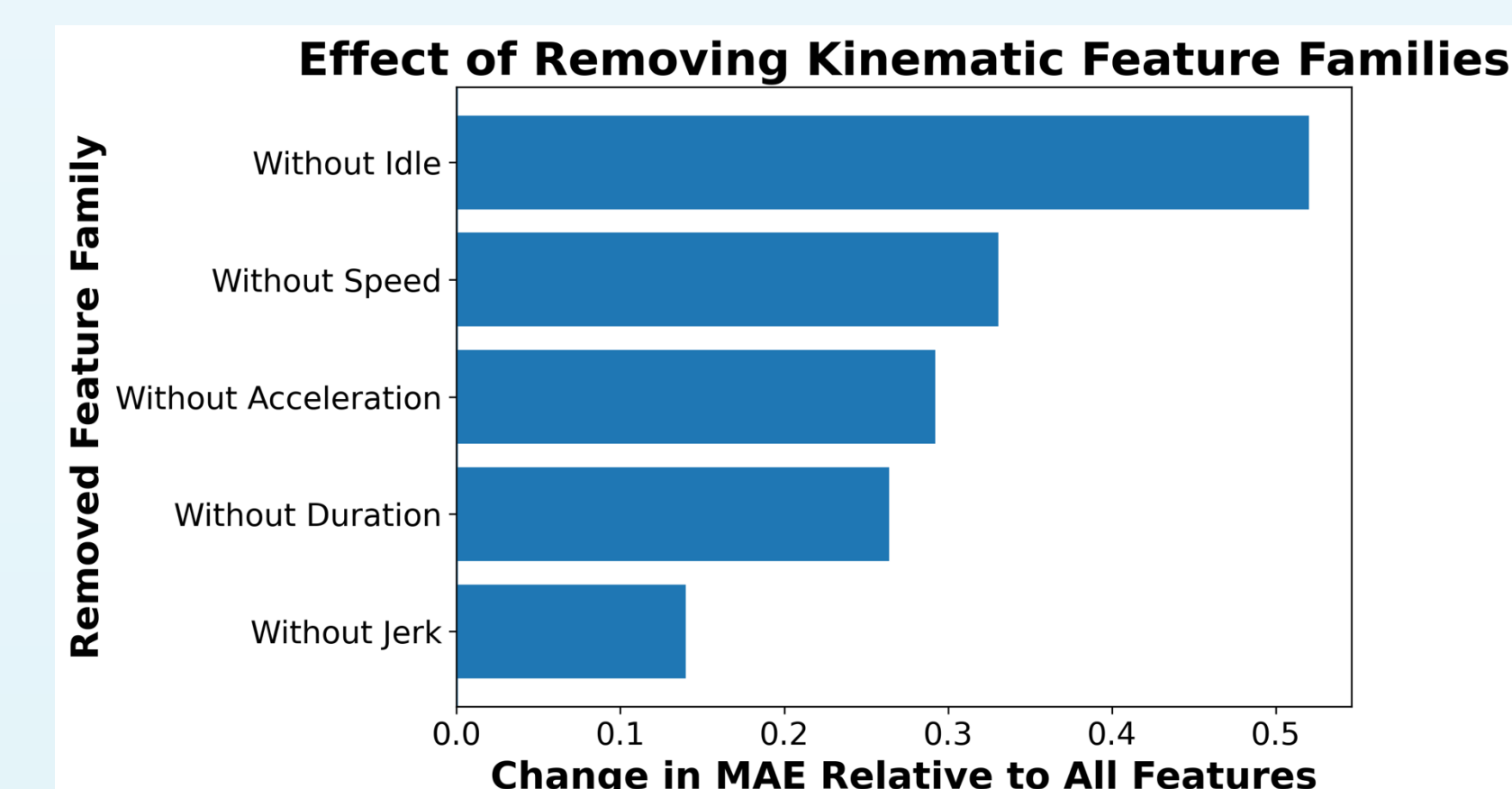
Feature-Family Ablation

Removing any feature family increased prediction error, showing that the model benefited from the combined representation.

MAE increase after feature removal:

- Idle: **+12.1%**
- Speed: **+7.7%**
- Acceleration: **+6.8%**
- Duration: **+6.1%**
- Jerk: **+3.3%**

Idle-related features provided the largest unique contribution.

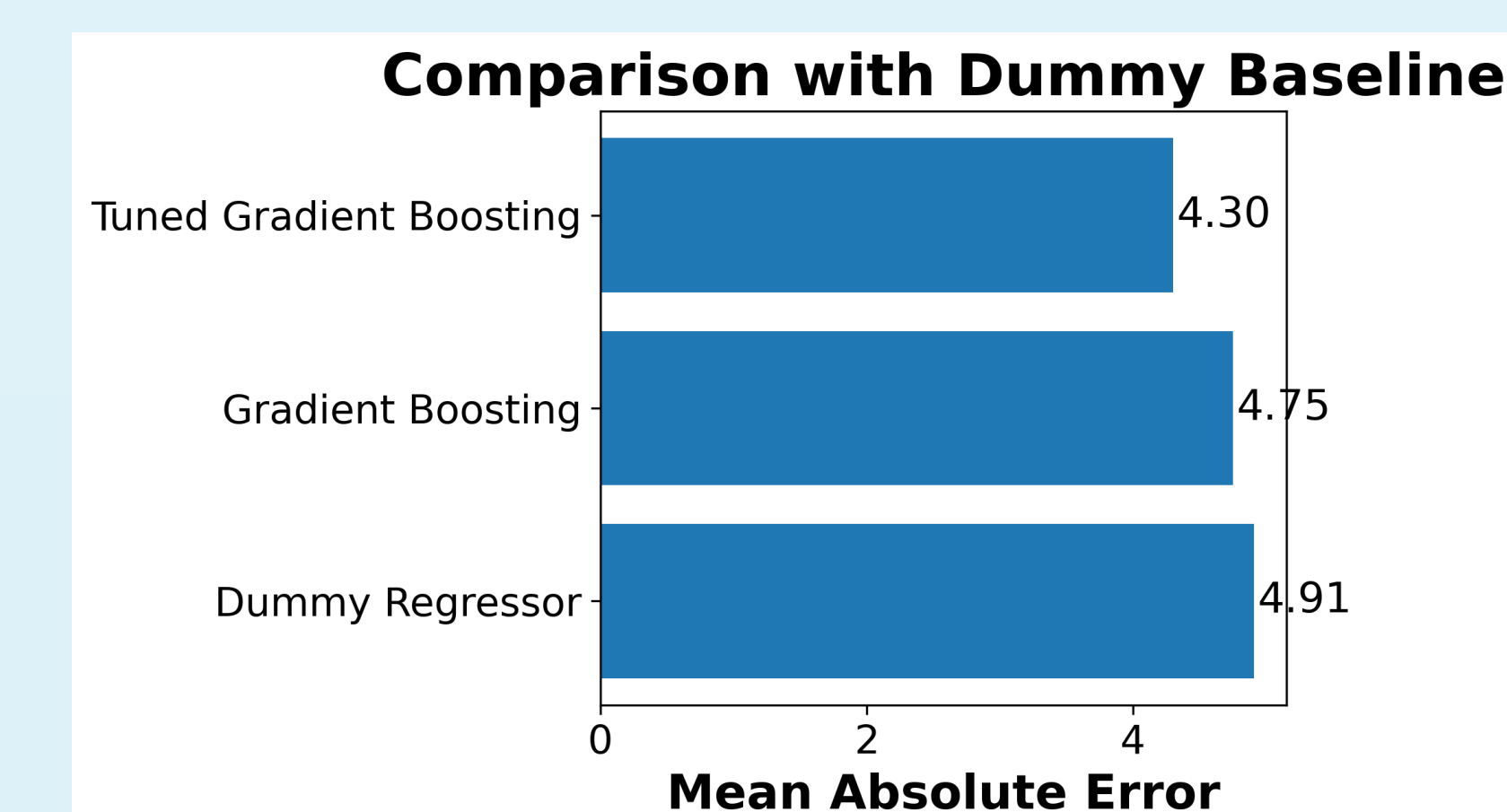


Baseline Comparison

Compared with simpler models:

Model	MAE
Dummy Regressor	4.91
Gradient Boosting	4.75
Tuned Gradient Boosting	4.30

Tuning reduced MAE by approximately 12% relative to the Dummy Regressor.



Discussion

The tuned Gradient Boosting model outperformed the baseline, indicating that the selected kinematic features contain measurable information related to skill. However, **performance remained modest**, with only 13% of GRS variation explained.

The model predicted middle-range scores more accurately than skill extremes. Lower scores were generally overestimated, while higher scores were underestimated, showing **regression toward the mean**.

Idle behavior, speed, and acceleration contributed most to performance. These results suggest that the current handcrafted feature set captures only part of the information associated with expertise.

Because evaluation used only surgical tasks, it remains **unknown whether these features are truly task-independent** or transferable to substantially different manipulation tasks.

Conclusion

- A tuned Gradient Boosting model achieved the best performance among the evaluated models (**MAE = 4.30**), outperforming baseline regression methods.
- Handcrafted task-independent kinematic features contain useful information related to human skill but do not fully explain variations in expert performance.
- Motion efficiency features, particularly idle time and speed, contributed most to predictive performance.
- The observed regression toward the mean indicates that the current feature representation is **insufficient for accurate continuous skill estimation**.
- These findings establish a baseline for evaluating more expressive representations of human motor skill in future work.

Future Work

Future work will investigate richer temporal and task-independent kinematic representations, test whether these features **generalize to non-surgical tasks** such as pick-and-place, and integrate the skill estimator into a **skill-based shared-autonomy system** that **adapts robotic assistance** according to estimated user skill.

Contributions

- Developed a continuous skill-estimation pipeline using robot kinematics
- Evaluated generalization using subject-wise cross-validation
- Quantified feature-family contributions through ablation
- Identified limitations of the current task-independent feature representation